UNIT-2: EIGEN VALUES AND EIGEN VECTORS

→ If 'A' is a square matrix of order 'n' then a vector 'x' is said to be Eigen vector if there exists a scalar 'λ' known as Eigen value

$$\Rightarrow \underbrace{AX}_{\text{Eigen vector}} = \underbrace{\lambda}_{\text{Eigen value}} X \rightarrow ①$$

Ex: $A = \begin{bmatrix} 5 & 4 \\ 2 & -1 \end{bmatrix}$ $X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$AX = \underset{A}{\begin{bmatrix} 5 & 4 \\ 2 & -1 \end{bmatrix}} \underset{X}{\begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \underset{\lambda}{1} \underset{X}{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}$$

① $\Rightarrow AX - \lambda X = 0$

$[A - \lambda I]X = 0$

$BX = 0 \rightarrow ②$ [where $B = A - \lambda I$)

→ Equation '2' denotes the system of homogeneous equations and will have a non-trival solution if

$\rho(B) < n \Rightarrow |B| = 0$

$\Rightarrow |A - \lambda I| = 0$, which is known as a characteristic equation which on expanding gives an n^{th} degree polynomial and will have 'n' roots known as Eigen values.

→ For each Eigen value we get Eigen vector by using $[A - \lambda I]X = 0$

Q) Find the Eigen values and Eigen vectors of (5, -2, 0) (-2, 6, 2) (0, 2, 7)

Sol)
$$\begin{bmatrix} 5 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 7 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The characteristics equation is $|A-\lambda I| = 0$

$$\begin{vmatrix} 5-\lambda & -2 & 0 \\ -2 & 6-\lambda & 2 \\ 0 & 2 & 7-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)\{(6-\lambda)(7-\lambda)-4\} - (-2)\{-2(7-\lambda)-0\} + 0 = 0$$

$\Rightarrow -\lambda^3 + (\text{sum of diagonal elements})\lambda^2 - \boxed{(\text{sum of cofactors of diagonal})\lambda}$ $+|A| = 0$

$$\Rightarrow -\lambda^3 + 18\lambda^2 - 99\lambda + 162 = 0$$

$$\lambda_1 = 3, \lambda_2 = 6, \lambda_3 = 9$$

which are eigen values

To find Eigen vectors:

For $\lambda_1 = 3$

$$(A-\lambda I)x = 0$$

$$\begin{bmatrix} 2 & -2 & 0 \\ -2 & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{array}{ccccccc} & x & & y & & z & \\ & & & & 2 & & -2 \\ -2 & & 0 & & & & \\ 2 & & 4 & & 0 & & 2 \end{array}$$

$$\frac{x}{-8+0} = \frac{y}{0-8} = \frac{z}{4-0}$$

$$\frac{x}{-8} = \frac{y}{-8} = \frac{z}{4}$$

$$\frac{x}{-2} = \frac{y}{-2} = \frac{z}{1}$$

$$X = \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 6$

$$\begin{bmatrix} -1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

$$\begin{array}{cccc} & x & y & z \\ -2 & 0 & -1 & -2 \\ 2 & 1 & 0 & 2 \end{array}$$

$$\frac{x}{-2-0} = \frac{y}{0+1} = \frac{z}{-2}$$

$$\frac{x}{-2} = \frac{y}{1} = \frac{z}{-2}$$

$$Y = \begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

for $\lambda_3 = 9$

$$\begin{bmatrix} -4 & -2 & 0 \\ -2 & -3 & 2 \\ 0 & 2 & -2 \end{bmatrix}$$

$$\begin{array}{cccc} & x & y & z \\ -2 & 0 & -4 & -2 \\ 2 & -2 & 0 & 2 \end{array}$$

$$\frac{x}{4-0} = \frac{y}{0-8} = \frac{z}{-8}$$

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$$\frac{x}{1} = \frac{y}{-2} = \frac{z}{-2}$$

$$X_3 = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \text{ or } \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$$

2) Find the Eigen values and Eigen vectors of

(1, -6, -4) (0, 4, 2) (0, -6, -3)

sol)
$$\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

The characteristic equation is $|A - \lambda I| = 0$

$$\begin{bmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{bmatrix}$$

$$-\lambda^3 + 2\lambda^2 - \lambda = 0$$

$$\lambda_1 = 1,\ \lambda_2 = 1,\ \lambda_3 = 0$$

To find Eigen vectors:

$\lambda = 0$

$$\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

$$\begin{array}{cccccccc} & x & & y & & z & & \\ -6 & & -4 & & 1 & & -6 & \\ 4 & & 2 & & 0 & & 4 & \end{array}$$

$$\frac{x}{-12+16} = \frac{y}{0-2} = \frac{z}{4-0}$$

$$\frac{x}{4} = \frac{y}{-2} = \frac{z}{4}$$

$$X = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$\lambda = 1$

$$\begin{bmatrix} 0 & -6 & -4 \\ 0 & 3 & 2 \\ 0 & -6 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \to 2R_2 + R_1$, $R_3 \to R_3 - R_1$

$$\begin{bmatrix} 0 & -6 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-6y - 4z = 0$$

$$3y = -2z$$

let $z = k_1, x = k_2$ then

$$y = \frac{-2}{3} k_1$$

$$z = k_1;\ x = k_2;\ y = \frac{-2}{3} k_1$$

$$X = \begin{bmatrix} k_2 \\ -\frac{2}{3} k_1 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} 0 \\ -2/3 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix} \quad X_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Diagonalization:

step 1: check whether the matrix is symmetric or not if not symmetric go to step 2.

step 2: Find Eigen values and eigen vectors.

step 3: construct modal matrix 'P' whose columns are eigen vectors of A (i.e $P = [X_1\ X_2\ X_3]$)

step 4: Find P inverse

step 5: Find $P^{-1}AP$

step 6: $P^{-1}AP = D$, which is a diagonal matrix whose diagonal elements are eigen values of A and the matrix is also known as spectral matrix

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→ using diagonalization we can find the powers of a matrix as follows

$\because P^{-1}AP = D$

$D[P^{-1}AP]D^{-1} = PDP^{-1}$

$PD^{-1}APD^{-1} = PDP^{-1}$

$IAI = PDP^{-1}$

$\boxed{A = PDD^{-1}}$

$\boxed{A^n = PD^nD^{-1}}$

Q) Diagonalise $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ also find A^5

sol) Find $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

characteristic matrix $|A-\lambda I| = 0$

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix}$$

$= -\lambda^3 + (-1)\lambda^2 - (-21)\lambda + 45$

$-\lambda^3 - \lambda^2 + 45 + 21\lambda = 0$

$\lambda = 5, -3, -3$

Eigen vectors:

$\lambda = -3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 2R_1$; $R_3 \rightarrow R_3 + R_1$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x + 2y - 3z = 0$

let $y = k_1$, $z = k_2$

$x = -2k_1 + 3k_2$

$$\begin{bmatrix} -2k_1 + 3k_2 \\ k_1 \\ k_2 \end{bmatrix} = k_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

$\lambda = 5$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & & z \\ 2 & & -3 & & -7 & 2 \\ -2 & & -5 & & -1 & -2 \end{array}$$

$$\frac{x}{-16} = \frac{y}{-32} = \frac{z}{16}$$

$$\frac{x}{-1} = \frac{y}{-2} = \frac{z}{1}$$

$$X = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} (\text{or}) \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \quad X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad X_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Modal matrix

$P = [X_1 \ \ X_2 \ \ X_3]$

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$$P = \begin{bmatrix} -1 & -2 & 3 \\ -2 & 1 & 0 \\ 10 & 0 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{\operatorname{adj} P}{|P|}$$

$$|P| = -8$$

$$\operatorname{adj} P : \begin{bmatrix} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix} \\ -\begin{vmatrix} -2 & 3 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 3 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} -1 & -2 \\ 1 & 0 \end{vmatrix} \\ \begin{vmatrix} -2 & 3 \\ 1 & 0 \end{vmatrix} & -\begin{vmatrix} -1 & 3 \\ -2 & 0 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ -2 & 1 \end{vmatrix} \end{bmatrix}^T$$

$$\operatorname{adj} P \begin{bmatrix} 1 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}$$

$$P^{-1} = \frac{1}{8}\begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ 1 & 2 & 5 \end{bmatrix}$$

$$P^{-1}AP = \frac{1}{8}\begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ 1 & 2 & 5 \end{bmatrix}\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}\begin{bmatrix} -1 & -2 & 3 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix} -5 & -10 & 15 \\ 6 & -12 & -18 \\ -3 & -6 & -15 \end{bmatrix}\begin{bmatrix} -1 & -2 & 3 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{5\times 6\times 3}{8}\begin{bmatrix} -1 & -2 & 3 \\ 1 & -2 & -3 \\ -1 & -2 & -5 \end{bmatrix}\begin{bmatrix} -1 & -2 & 3 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \frac{90}{8}\begin{bmatrix} 8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -8 \end{bmatrix}$$

$$= \frac{4\times 90}{8}\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} = 45\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

To find A^5:

we know that $A^n = PD^nP^{-1}$

$$A^5 = PD^5D^{-1}$$

$$= \frac{1}{8}\begin{bmatrix}-1 & -2 & 3\\ -2 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}\begin{bmatrix}5^5 & 0 & 0\\ 0 & -3^5 & 0\\ 0 & 0 & -3^5\end{bmatrix}\begin{bmatrix}-1 & -2 & 3\\ -2 & 4 & 6\\ 1 & 2 & 5\end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix}-1 & -2 & 3\\ -2 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}\begin{bmatrix}3125 & 0 & 0\\ 0 & -243 & 0\\ 0 & 0 & -243\end{bmatrix}\begin{bmatrix}-1 & -2 & 3\\ -2 & 4 & 6\\ 1 & 2 & 5\end{bmatrix}$$

$$= \frac{1}{+8}\begin{bmatrix}-3125 & 486 & -729\\ 6250 & -243 & 0\\ 3125 & 0 & -243\end{bmatrix}\begin{bmatrix}-1 & -2 & 3\\ -2 & 4 & 6\\ 1 & 2 & 5\end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix}1424 & 6736 & -10104\\ 6736 & 11528 & -20208\\ -3368 & -6736 & 8166\end{bmatrix} = \begin{bmatrix}178 & 842 & -1263\\ 842 & 1441 & -2526\\ -421 & -842 & 1020\end{bmatrix}$$

Diagonalisation of real symmetric matrix (orthogonal transformation)

step1: check whether the matrix is symmetric or not if so go to step 2

step2: Find Eigen values and Eigen vectors

step 3: check whether the eigen vectors are orthogonal or not (i.e $x_1^T \cdot x_j = 0$)

→ If the vectors are not orthogonal then replace that vectors which is not orthogonal with the new vector.

step4: construct the normalized modern matrix.

$$P = \left[\frac{x_1}{\|x_1\|}, \frac{x_2}{\|x_2\|}, \frac{x_3}{\|x_3\|}\right]$$

If $x_i = \begin{bmatrix}a\\ b\\ c\end{bmatrix}$ then $\|x_i\| = \sqrt{a^2+b^2+c^2}$

step 5: Find $P^TAP = D$

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Q) Diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Sol) The characteristic equation $|A-\lambda I| = 0$

$$\begin{bmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix}$$

$$\Rightarrow -\lambda^3 + 11\lambda^2 - 36\lambda + 36 = 0$$

$$\lambda_1 = 6, \lambda_2 = 3, \lambda_3 = 2$$

for $\lambda_1 = 2$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & & z \\ 3 & & -1 & & -1 & & 3 \\ -1 & & 1 & & 1 & & -1 \end{array}$$

$$\frac{x}{3-1} = \frac{y}{-1+1} = \frac{z}{1-3}$$

$$\frac{x}{2} = \frac{y}{0} = \frac{z}{-2}$$

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$$

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

for $\lambda_2 = 3$

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & & z \\ 2 & & -1 & & -1 & & 2 \\ -1 & & 0 & & 1 & & -1 \end{array}$$

$$\frac{x}{0-1} = \frac{y}{-1+0} = \frac{z}{1-2}$$

$$\frac{x}{-1} = \frac{y}{-1} = \frac{z}{-1}$$

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$$X_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for $\lambda_3 = 6$

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

$$\begin{array}{cccccccc} & x & & y & & z & & \\ -1 & & -1 & & -1 & & -1 \\ -1 & & -3 & & 1 & & -1 \end{array}$$

$$\frac{x}{3-1} = \frac{y}{-1-3} = \frac{z}{1+1}$$

$$\frac{x}{2} = \frac{y}{-4} = \frac{z}{2}.$$

$$X = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad X_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

As the given matrix is symmetric check the orthogonality of 3 vectors.

$$X_1^T \cdot X_2 = [1 \quad -2 \quad 1] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0.$$

$$X_2^T \cdot X_3 = [1 \quad 1 \quad 1] \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$X_3^T \cdot X_1 = [-1 \quad 0 \quad 1] \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = 0$$

X_1, X_2, X_3 are pair wise orthogonal

$$\|X_1\| = \sqrt{(1)^2+(-2)^2+(1)^2} = \sqrt{6}$$

$$\|X_2\| = \sqrt{1^2+1^2+1^2} = \sqrt{3}$$

$$\|X_3\| = \sqrt{(-1)^2+0^2+1^2} = \sqrt{2}$$

Normalized modern matrix

$$P = \begin{bmatrix} \frac{X_1}{\|X_1\|} & \frac{X_2}{\|X_2\|} & \frac{X_3}{\|X_3\|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & -\frac{1}{\sqrt2} \\ -\frac{2}{\sqrt6} & \frac{1}{\sqrt3} & \frac{0}{\sqrt2} \\ -\frac{1}{\sqrt6} & \frac{1}{\sqrt3} & \frac{1}{\sqrt2} \end{bmatrix}$$

$$P^T = \begin{bmatrix} \frac{1}{\sqrt6} & -\frac{2}{\sqrt6} & -\frac{1}{\sqrt6} \\ \frac{1}{\sqrt3} & \frac{1}{\sqrt3} & \frac{1}{\sqrt3} \\ -\frac{1}{\sqrt2} & \frac{0}{\sqrt2} & \frac{1}{\sqrt2} \end{bmatrix}$$

$$P^TAP = \begin{bmatrix} \frac{1}{\sqrt6} & -\frac{2}{\sqrt6} & -\frac{1}{\sqrt6} \\ \frac{1}{\sqrt3} & \frac{1}{\sqrt3} & \frac{1}{\sqrt3} \\ -\frac{1}{\sqrt2} & \frac{0}{\sqrt2} & \frac{1}{\sqrt2} \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 6 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & -\frac{1}{\sqrt2} \\ -\frac{2}{\sqrt6} & \frac{1}{\sqrt3} & 0 \\ \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & \frac{1}{\sqrt2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3+2+1}{\sqrt6} & \frac{-1-10-1}{\sqrt6} & \frac{1+2+3}{\sqrt6} \\ \frac{3-1+1}{\sqrt3} & \frac{-1+5-1}{\sqrt3} & \frac{1-1+3}{\sqrt3} \\ \frac{-3+0+1}{\sqrt2} & \frac{1+0-1}{\sqrt2} & \frac{-1+0+3}{\sqrt2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & -\frac{1}{\sqrt2} \\ -\frac{2}{\sqrt6} & \frac{1}{\sqrt3} & 0 \\ \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & \frac{1}{\sqrt2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{6}{\sqrt6} & \frac{-12}{\sqrt6} & \frac{6}{\sqrt6} \\ \frac{3}{\sqrt3} & \frac{3}{\sqrt3} & \frac{3}{\sqrt3} \\ -\frac{2}{\sqrt2} & 0 & \frac{2}{\sqrt2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & -\frac{1}{\sqrt2} \\ -\frac{2}{\sqrt6} & \frac{1}{\sqrt3} & 0 \\ \frac{1}{\sqrt6} & \frac{1}{\sqrt3} & \frac{1}{\sqrt2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{6+24+6}{6} & \frac{6-12+6}{\sqrt{18}} & -\frac{6+0+6}{\sqrt{12}} \\ \frac{3+3+3}{\sqrt{18}} & \frac{3+3+3}{3} & \frac{-3+0+3}{\sqrt6} \\ \frac{-2+0+2}{\sqrt{12}} & \frac{-2+0+2}{\sqrt6} & \frac{2+0+2}{2} \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Q) Diagonalize $\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

sol) characteristic matrix $|A-\lambda I|=0$

$$\begin{bmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{bmatrix} = 0$$

$$-\lambda^3+6\lambda^2-9\lambda = 0$$

$$\lambda = 0,3,3$$

$\lambda_1=0$

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \qquad \begin{array}{cccccccc} & x & & y & & z & \\ 2 & & -1 & & -1 & & 2 \\ -1 & & 2 & & -1 & & -1 \end{array}$$

$$\frac{x}{4-1} = \frac{y}{1+2} = \frac{z}{1+2}$$

$$\frac{x}{3} = \frac{y}{3} = \frac{z}{3}$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\lambda_2 = 3$

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-x-y-z=0$$

$$x+y+z=0$$

let $y=k_1, z=k_2$

$$x+k_1+k_2=0$$

$$x=-(k_1+k_2)$$

$$X=\begin{bmatrix}-k_1-k_2\\k_1\\k_2\end{bmatrix}=\begin{bmatrix}-k_1\\k_1\\0\end{bmatrix}+\begin{bmatrix}-k_2\\0\\k_2\end{bmatrix}$$

$$k_1\begin{bmatrix}-1\\1\\0\end{bmatrix}+k_2\begin{bmatrix}-1\\0\\1\end{bmatrix}$$

$$X_2=\begin{bmatrix}-1\\1\\0\end{bmatrix}\quad X_3\begin{bmatrix}-1\\0\\1\end{bmatrix}$$

$$X_1=\begin{bmatrix}1\\1\\1\end{bmatrix}\quad X_2=\begin{bmatrix}-1\\1\\0\end{bmatrix}\quad X_3=\begin{bmatrix}-1\\0\\1\end{bmatrix}$$

$$X_1^T X_2=\begin{bmatrix}1&1&1\end{bmatrix}\begin{bmatrix}-1\\1\\0\end{bmatrix}=0$$

$$X_2^T X_3=\begin{bmatrix}-1&1&0\end{bmatrix}\begin{bmatrix}-1\\0\\1\end{bmatrix}=1\neq 0$$

$$X_3^T X_1=\begin{bmatrix}-1&0&1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}=0$$

Here orthogonality fails for X_2 and X_3 so replace either X_2 or X_3 by a new vector which is orthogonal with remaining vectors

let $X_1=\begin{bmatrix}1\\1\\1\end{bmatrix}\quad X_2=\begin{bmatrix}a\\b\\c\end{bmatrix}\quad X_3=\begin{bmatrix}-1\\0\\1\end{bmatrix}$

$$X_1^T X_2=0 \Rightarrow a+b+c=0 \rightarrow (1)$$

$$X_2^T X_3=0 \Rightarrow -a+0+c=0 \rightarrow (2)$$

solve eq^n (1) + (2)

$$\begin{array}{cccccc} & a & & b & & c \\ 1 & & 1 & & 1 & & 1 \\ 0 & & 1 & & -1 & & 0 \end{array}$$

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$$\frac{a}{1-0} = \frac{b}{-1-1} = \frac{c}{0+1} \Rightarrow \frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$$

$$X_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad X_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\|X_1\| = \sqrt{1^2+1^2+1^2} = \sqrt{3}; \ \|X_2\| = \sqrt{1+4+1} = \sqrt{6}; \ \|X_3\| = \sqrt{1+1} = \sqrt{2}$$

Normalised modal matrix P

$$P = \begin{bmatrix} \frac{X_1}{\|X_1\|} & \frac{X_2}{\|X_2\|} & \frac{X_3}{\|X_3\|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$P^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$P^TAP = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2-1-1}{\sqrt{3}} & \frac{-1+2-1}{\sqrt{3}} & \frac{-1-1+2}{\sqrt{3}} \\ \frac{2+2-1}{\sqrt{6}} & \frac{-1-4-1}{\sqrt{6}} & \frac{-1+2+2}{\sqrt{6}} \\ \frac{-2+0-1}{\sqrt{2}} & \frac{1+0-1}{\sqrt{2}} & \frac{1+0+2}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ \frac{3}{\sqrt{6}} & \frac{-6}{\sqrt{6}} & \frac{3}{\sqrt{6}} \\ \frac{-3}{\sqrt{2}} & 0 & \frac{3}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ \frac{3-6+3}{\sqrt{18}} & \frac{3+12+3}{6} & \frac{-3+0+3}{\sqrt{12}} \\ \frac{-3+0+3}{\sqrt{6}} & \frac{-3+0+3}{\sqrt{12}} & \frac{3+0+3}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{18}{6} & 0 \\ 0 & 0 & \frac{6}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Quadratic Form:

A homogeneous expression of degree 2 is known as a quadratic expression

→ An equation of the form $Q = X^TAX = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_i x_j$ is known as quadratic form in n variables where 'A' is a real symmetric matrix known as the matrix of quadratic form

Q) Write the symmetric matrix A associated with the given quadratic form $2x^2+3y^2+z^2-5xy+7yz+2zx$

Sol) $X^TAX \Rightarrow [x \quad y \quad z] \begin{bmatrix} 2 & -5/2 & 1 \\ -5/2 & 3 & 7/2 \\ 1 & 7/2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Q) Write the quadratic form of given matrix

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 1 & 7 \\ 3 & 7 & 9 \end{bmatrix}$$

Sol) $Q = X^TAX$

$[x \quad y \quad z] \begin{bmatrix} 5 & 2 & 3 \\ 2 & 1 & 7 \\ 3 & 7 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$5x^2+y^2+9z^2+4xy+6xz+14yz$

Rank of the quadratic form:

Rank of real symmetric matrix 'A' associated with the given quadratic form is the rank of quadratic form.

→ The number of non zero Eigen values gives the rank of A

canonical form:

let $Q = X^T A X$ be the quadratic form let $X = PY$ be the transformation which transforms the given quadratic form to another quadratic form which is known canonical form (or) normal form (or) sum of squares form

(i.e) $Q = X^T A X$ $\quad [(AB)^T = B^T A^T]$

$(PY)^T A (PY)$

$Y^T P^T A P Y$

$Y^T (P^T A P) Y$

$Y^T D Y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$

where $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ are eigen values of A

Index of quadratic form:

The number of positive eigen values is known as index denoted by 's'

signature:

→ Excess of positive terms over the negative terms is known as signature

$$sgn = 2s - r$$

Ex: $-1, 2, 3$ are eigen values

$s = 2$

$r = 3$

$sgn = 1$

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Nature of quadratic form:

1) positive definite:

If all eigen values are positive then quadratic form is said to be positive definite (i.e $\Rightarrow s = r = n$)

2) positive semi definite:

If atleast one eigen value is zero and the remaining are positive then it is known as positive semi definite

i.e $r < n$ & $s = r$

3) Negative definite:

If all eigen values are negative then it is negative definite

(i.e $r = n$ & $s = 0$)

4) Negative semi definite:

If atleast one eigen value is zero and the remaining are negative then it is negative semi definite

i.e $r < n$ & $s = 0$

5) Indefinite:

If some eigen values are positive and remaining are negative then it is indefinite.

Q) Reduce the quadratic form to canonical form and find the nature, index and signature

$$3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$$

sol)

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

The matrix of quadratic form is given by A

The characteristic equation $|A-\lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix}$$

$$\Rightarrow -\lambda^3+11\lambda^2-36\lambda+36=0$$

$$\lambda_1=6,\ \lambda_2=3,\ \lambda_3=2$$

for $\lambda=6$

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & z & \\ -1 & & -1 & & -1 & -1 \\ -1 & & -3 & & 1 & -1 \end{array}$$

$$\frac{x}{3-1} = \frac{y}{-1-3} = \frac{z}{1+1}$$

$$\frac{x}{2} = \frac{y}{-4} = \frac{z}{2}$$

$$X_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

for $\lambda_2 = 3$

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & z & \\ -1 & & 1 & & 0 & -1 \\ 2 & & -1 & & -1 & 2 \end{array}$$

$$\frac{x}{1-2} = \frac{y}{-1-0} = \frac{z}{0-1}$$

$$\frac{x}{-1} = \frac{y}{-1} = \frac{z}{-1}$$

$$X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for $\lambda = 2$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\begin{array}{cccccc} & x & & y & z & \\ 3 & & -1 & -1 & & 3 \\ -1 & & 1 & 1 & & -1 \end{array}$$

$$\frac{x}{3-1} = \frac{y}{-1+1} = \frac{z}{1-3}$$

$$\frac{x}{2} = \frac{y}{0} = \frac{z}{-2}$$

$$X_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

orthogonality:

$$X_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad X_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$X_1^T X_2 = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$$

$$X_2^T X_3 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 0$$

$$X_3^T X_1 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = 0$$

Normalised matrix:

$\|X_1\| = \sqrt{1+4+1} = \sqrt{6}$; $\|X_2\| = \sqrt{3}$; $\|X_3\| = \sqrt{2}$

$$P \begin{bmatrix} \frac{X_1}{\|X_1\|} & \frac{X_2}{\|X_2\|} & \frac{X_3}{\|X_3\|} \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$$P^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$P^TAP = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3+2+1}{\sqrt{6}} & \frac{-1-10-1}{\sqrt{6}} & \frac{1+2+3}{\sqrt{6}} \\ \frac{3-1+1}{\sqrt{3}} & \frac{-1+5-1}{\sqrt{3}} & \frac{1-1+3}{\sqrt{3}} \\ \frac{3+0-1}{\sqrt{2}} & \frac{-1+0+1}{\sqrt{2}} & \frac{1+0-3}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{6}{\sqrt{6}} & \frac{-12}{\sqrt{6}} & \frac{6}{\sqrt{6}} \\ \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} \\ \frac{2}{\sqrt{2}} & 0 & \frac{-2}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{6+24+6}{6} & \frac{6-12+6}{\sqrt{18}} & \frac{6+0-6}{\sqrt{12}} \\ \frac{3-6+3}{\sqrt{18}} & \frac{3+3+3}{3} & \frac{3+0-3}{\sqrt{6}} \\ \frac{2+0-2}{\sqrt{12}} & \frac{2+0-2}{\sqrt{6}} & \frac{2+0+2}{2} \end{bmatrix}$$

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$$= \begin{bmatrix} 36/6 & 0 & 0 \\ 0 & 9/3 & 0 \\ 0 & 0 & 4/2 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

canonical form is $Y^T D Y$

$$[y_1 \ y_2 \ y_3] \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$6y_1^2 + 3y_2^2 + 2y_3^2$

rank $= 3$

$s = 3$

$sg = 2s - r = 3$

Nature $\Rightarrow$ positive definite

cayley Hamilton theorem:

Every square matrix satisfies it characteristic equation.

Application of cayley hamilton theorem:

→ using cayley hamilton theorem we can find

i) inverse of a matrix

ii) Highest power of a matrix

Q verify cayley Hamilton theorem and A^{-1} and A^5

$$A = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 2 \\ 2 & 3 & 4 \end{bmatrix}$$

sol) The characteristic equation is $|A-\lambda I|=0$

$$\begin{vmatrix} -1-\lambda & -2 & 0 \\ 1 & -\lambda & 2 \\ 2 & 3 & 4-\lambda \end{vmatrix}$$

$$-\lambda^3+3\lambda^2+8\lambda+6=0$$

verification:

we have to prove that $-A^3+3A^2+8A+6I=0$

$$A^2 = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 2 \\ 2 & 3 & 4 \end{bmatrix}\begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 2 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 2 & -4 \\ 3 & 4 & 8 \\ 9 & 8 & 22 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} -1 & 2 & -4 \\ 3 & 4 & 8 \\ 9 & 8 & 22 \end{bmatrix}\begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 2 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & -10 & -12 \\ 17 & 18 & 40 \\ 43 & 48 & 104 \end{bmatrix}$$

$$-A^3+3A^2+8A+6I=0$$

$$\begin{bmatrix} 5 & 10 & 12 \\ -17 & -18 & -40 \\ -43 & -48 & -104 \end{bmatrix} + 3\begin{bmatrix} -1 & 2 & -4 \\ 3 & 4 & 8 \\ 9 & 8 & 22 \end{bmatrix} + 8\begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 2 \\ 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 10 & 12 \\ -17 & -18 & -40 \\ -43 & -48 & -104 \end{bmatrix} + \begin{bmatrix} -3 & 6 & -12 \\ 9 & 12 & 24 \\ 27 & 24 & 60 \end{bmatrix} + \begin{bmatrix} -8 & -16 & 0 \\ 8 & 0 & 16 \\ 16 & 24 & 32 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 5-3-8+6 & 10+6-6+0 & 12-12+0+0 \\ -17+9+8+0 & -18+12+0+6 & -40+24+16+0 \\ -43+27+16+0 & -48+24+24+0 & -104+36+60+6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Find A^{-1}

since $A^3-3A^2-8A-6I=0$

multiply with A^{-1} on b.s we get

$$A^2-3A-8I-6A^{-1}=0$$

$$6A^{-1}=A^2-3A-8I$$

$$6A^{-1}=\begin{bmatrix}-1 & 2 & -4\\ 3 & 4 & 8\\ 9 & 8 & 22\end{bmatrix}-3\begin{bmatrix}-1 & -2 & 0\\ 1 & 0 & 2\\ 2 & 3 & 4\end{bmatrix}-8\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}-1 & 2 & -4\\ 3 & 4 & 8\\ 9 & 8 & 22\end{bmatrix}-\begin{bmatrix}-3 & -6 & 0\\ 3 & 0 & 6\\ 6 & 9 & 12\end{bmatrix}-\begin{bmatrix}8 & 0 & 0\\ 0 & 8 & 0\\ 0 & 0 & 8\end{bmatrix}$$

$$=\begin{bmatrix}-1+3-8 & 2+6+0 & -4-0-0\\ 3-3-0 & 4-0-8 & 8-6-0\\ 9-6-0 & 8-9-0 & 22-12-8\end{bmatrix}$$

$$6A^{-1}=\begin{bmatrix}-6 & 8 & -4\\ 0 & -4 & 2\\ 3 & -1 & 2\end{bmatrix}$$

$$A^{-1}=\frac{1}{6}\begin{bmatrix}-6 & 8 & -4\\ 0 & -4 & 2\\ 3 & -1 & 2\end{bmatrix}$$

To find A^4

$$A^3-3A^2-8A-6I=0$$

$$A^4-3A^3-8A^2-6A=0$$

$$A^4=3A^3+8A^2+6A$$

$$A^4=3\begin{bmatrix}-5 & -10 & -12\\ 17 & 18 & 40\\ 43 & 48 & 104\end{bmatrix}+8\begin{bmatrix}-1 & 2 & -4\\ 3 & 4 & 8\\ 9 & 8 & 22\end{bmatrix}+6\begin{bmatrix}-1 & -2 & 0\\ 1 & 0 & 2\\ 2 & 3 & 4\end{bmatrix}$$

$$=\begin{bmatrix}-15 & -30 & -36\\ 51 & 54 & 120\\ 129 & 144 & 312\end{bmatrix}+\begin{bmatrix}-8 & 16 & -32\\ 24 & 32 & 64\\ 72 & 64 & 116\end{bmatrix}+\begin{bmatrix}-6 & -12 & 0\\ 6 & 0 & 12\\ 12 & 18 & 24\end{bmatrix}$$

$$= \begin{bmatrix} -15-8-6 & -30+16-12 & -36-32+0 \\ 51+24+6 & 54+32+0 & 120+64+12 \\ 129+72+12 & 144+64+18 & 312+176+24 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} -29 & -26 & -68 \\ 81 & 86 & 196 \\ 213 & 226 & 512 \end{bmatrix}.$$

Q) $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ find the value of the matrix

$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ and verify cayley Hamilton theorem

sol) $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

The characteristic equation $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix}$$

$$-\lambda^3 + 5\lambda^2 - 7\lambda + 3 = 0$$

P.T $-A^3 + 5A^2 - 7A + 3I = 0$

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$-A^3 + 5A^2 - 7A + 3I = 0$$

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$$\begin{bmatrix} -14 & -13 & -13 \\ 0 & -1 & 0 \\ -13 & -13 & -14 \end{bmatrix} + 5\begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 7\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -14 & -13 & -13 \\ 0 & -1 & 0 \\ -13 & -13 & -14 \end{bmatrix} + \begin{bmatrix} 25 & 20 & 20 \\ 0 & 5 & 0 \\ 20 & 20 & 25 \end{bmatrix} - \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -14+25-14+3 & -13+20-7+0 & -13+20+7+0 \\ 0+0+0+0 & -1+5-7+3 & 0+0+0+0 \\ -13+20-7+0 & -13+20-7 & -14+25-14+3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

consider $A^8-5A^7+7A^6-3A^5+A^4-5A^3+8A^2-2A+I$

$A^5[A^3-5A^2+7A-3I]+A^4-5A^3+8A^2-2A+I$

$A[A^3-5A^2+8A-2I]+I$

$A[A^3-5A^2+7A+A-3I+I]+I$

$A[(A^3-5A^2+7A-3I)+A+I)+I$

$A(A+I)+I$

A^2+A+I

$$\begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5+2+1 & 4+1+0 & 4+1+0 \\ 0+0+0 & 1+1+1 & 0+0+0 \\ 4+1+0 & 4+1+0 & 5+2+1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

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Prove that sum of Eigen values is equal to trace and product of Eigen value is the determinant.

A: consider $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

The characteristic equation is $|AI - \lambda| = 0$

$$A = \begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} = 0$$

$(a_{11}-\lambda)\left((a_{22}-\lambda)(a_{33}-\lambda) - a_{23}a_{32}\right) - a_{12}\left[a_{21}(a_{33}-\lambda) - a_{31}a_{23}\right] +$

$a_{13}\left[a_{21}a_{32} - a_{31}(a_{22}-\lambda)\right] = 0$

$(a_{11}-\lambda)(a_{22}-\lambda)(a_{33}-\lambda) - (a_{11}-\lambda)a_{23}a_{32} - a_{12}a_{21}(a_{33}-\lambda) + a_{12}a_{31}a_{23} +$

$a_{13}a_{21}a_{32} - a_{31}a_{13}(a_{22}-\lambda) = 0$

$(a_{11}a_{22} - a_{11}\lambda - \lambda a_{22} + \lambda^2)(a_{33}-\lambda) - (a_{11}-\lambda)a_{23}a_{32} - a_{12}a_{21}(a_{33}-\lambda) +$

$a_{12}a_{31}a_{23} + a_{13}a_{21}a_{32} - a_{31}a_{13}(a_{22}-\lambda) = 0$

$a_{11}a_{33}a_{22} - a_{11}a_{33}\lambda - \lambda a_{22}a_{33} + \lambda^2 a_{33} - \lambda a_{11}a_{22} + a_{11}\lambda^2 + \lambda^2 a_{22}$

$-\lambda^3 - a_{11}a_{23}a_{32} + \lambda a_{23}a_{32} - a_{12}a_{21}a_{33} + \lambda a_{12}a_{21} + a_{12}a_{31}a_{23} + a_{13}a_{21}a_{32}$

$-a_{31}a_{13}a_{22} + \lambda a_{31}a_{13} = 0$

$-\lambda^3 + \lambda^2(a_{33} + a_{11} + a_{22}) - \lambda(a_{11}a_{33} + a_{22}a_{33} + a_{11}a_{22} - a_{23}a_{32} - a_{12}a_{21} - a_{31}a_{13})$

$+ a_{11}a_{33}a_{22} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{31}a_{23} + a_{13}a_{21}a_{32} -$

$a_{31}a_{13}a_{22} = 0$

$-\lambda^3 + (a_{33} + a_{11} + a_{22})\lambda^2 - \lambda(a_{11}a_{22} + a_{33}a_{22} + a_{11}a_{33} - a_{23}a_{32} - a_{12}a_{21} -$

$a_{31}a_{13}) + \{a_{11}(a_{33}a_{22} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} -$

$a_{31}a_{22})\} = 0$

which is a cubic equation

let $\lambda_1, \lambda_2, \lambda_3$ be the characteristic roots of the equation

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since sum of the roots $= -\frac{b}{a} = \frac{+(a_{11}+a_{22}+a_{33})}{+1}$

$$\lambda_1+\lambda_2+\lambda_3 = a_{11}+a_{22}+a_{33}$$
$$= \text{Trace of } A$$

and the product of roots $= \frac{-d}{a}$

$$\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = \frac{-|A|}{-1} = |A|$$

2) If λ is eigen value of A and x is the eigen vector then λ^n is the eigen value of A^n corresponding to the eigen vector x

Proof: since λ is eigen value and x is the eigen vector

$$AX = \lambda X \rightarrow ①$$
multiply by A
$$A(AX) = (\lambda X)A$$
$$A^2X = (AX)\lambda$$
$$A^2X = (\lambda X)\lambda$$
$$A^2X = \lambda^2 X$$

$\rightarrow \lambda^2$ is eigen value of A^2 similarly again multiply with A on both sides we get $A^3X = \lambda^3 X$

$\rightarrow \lambda^3$ is eigen value of A^3 hence λ^n is eigen value of A^n

3) If $\lambda_1, \lambda_2, --- \lambda_n$ are eigen values (or) latent roots of A then $\lambda_1^n, \lambda_2^n, \lambda_3^n, --- \lambda_n^n$ are eigen values of A^n

4) The square matrix A and A^T have same eigen values.

The characteristic equation of A is $|A-\lambda I|$

consider $(A-\lambda I)^T = (A^T - \lambda I^T) = (A^T - \lambda I)$

Taking Det on both sides

$$|(A-\lambda I)^T| = |A^T - \lambda I| \qquad (\because |B^T| = |B|)$$
$$|A-\lambda I| = |A^T - \lambda I|$$

$\rightarrow$ characteristic equation of A = characteristic equation of A^T

$\rightarrow$ Eigen values of A = Eigen values of A^T

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5) If $\lambda_1, \lambda_2, \lambda_3 ---- \lambda_n$ are eigen values of A then $k\lambda_1, k\lambda_2, k\lambda_3 --- k\lambda_n$ are eigen values of KA

6) If x is an eigen vector of A then x cannot correspond to more than one eigen value.

Proof: let λ_1, λ_2 be the two eigen values corresponding to the same eigen vector x then $AX = \lambda_1 X$ and $AX = \lambda_2 X$

$$\lambda_1 X = \lambda_2 X$$
$$(\lambda_1 - \lambda_2) X = 0$$
$$\lambda_1 - \lambda_2 = 0$$
$$\lambda_1 = \lambda_2$$

$\therefore$ x cannot correspond to more than one eigen value.

7) If λ is eigen value of A then $\lambda + k$ is eigen value of $A + KI$

8) If $\lambda_1, \lambda_2, \lambda_3 --- \lambda_n$ are eigen values of A then $\lambda_1 - k, \lambda_2 - k, -- \lambda_n - k$ are eigen values of $A - KI$.

9) If λ is eigen value of A then λ^{-1} (or) $\frac{1}{\lambda}$ is eigen value of A^{-1}

Proof: If λ is eigen value of A then

$$AX = \lambda X$$
$$A^{-1}(AX) = A^{-1}(\lambda X)$$
$$X = A^{-1} \lambda X$$
$$\frac{1}{\lambda} X = A^{-1} X$$
$$A^{-1} X = \frac{1}{\lambda} X$$

$\frac{1}{\lambda}$ is eigen vale of A^{-1}

10) If λ is eigen value of a non singular matrix A then $\frac{|A|}{\lambda}$ is eigen value of adj A

Proof: since λ is eigen value and x is eigen vector

$$AX = \lambda X$$

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multiply by adj a on b.s

$(adj\,A)(A\lambda) = \lambda\,(adj\,A)X \rightarrow ①$

w.k.T $\quad A^{-1} = \dfrac{adj\,A}{|A|}$

$A^{-1}A = \dfrac{adj\,A \cdot A}{|A|}$

$I = \dfrac{(adj\,A)}{|A|} \cdot A$

$|A| = (adj\,A) \cdot A$

eq ① $\Rightarrow \quad |A|\,X = \lambda\,(adj\,A)\,X$

$(adj\,A)X = \dfrac{|A|}{\lambda}X$

$\dfrac{|A|}{\lambda}$ is eigen value of adj A

ii) If λ is eigen value of an orthogonal matrix then $\frac{1}{\lambda}$ is also its eigen value.

Proof: If A is an orthogonal matrix then $\boxed{A \cdot A^T = I}$

$A^T = A^{-1}$

we know that If λ is eigen value of A

$\lambda \qquad A$

$\dfrac{1}{\lambda} \qquad A^{-1}$

$\frac{1}{\lambda}$ is eigen value of A^T $\quad (\because A^{-1} = A^T)$

$\frac{1}{\lambda}$ is eigen value of A $\quad (\because A$ & A^T have same eigen values)

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12) If λ is eigen value of A then prove that eigen value of $B = a_0A^2 + a_1A + a_2I$ is $a_0\lambda^2 + a_1\lambda + a_2$

Proof:

If λ is eigen value & X is eigen vector

$$AX = \lambda X$$

$$A^2X = \lambda^2 X$$

consider $B = a_0A^2 + a_1A + a_2I$ then

$$BX = a_0A^2X + a_1AX + a_2X$$

$$a_0\lambda^2 + a_2\lambda + a_2X$$

$$BX = [a_0\lambda^2 + a_1\lambda + a_2]X$$

$a_0\lambda^2 + a_1\lambda + a_2$ is eigen value of X

3) The eigen values of a triangular matrix are diagonal elements

Proof: consider an upper triangular matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

The characteristic equation is $|A - \lambda I| = 0$

$$\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ 0 & a_{21}-\lambda & a_{23} \\ 0 & 0 & a_{33}-\lambda \end{vmatrix}$$

$$(a_{11}-\lambda)\{(a_{22}-\lambda)(a_{33}-\lambda)\} - 0 + 0 = 0$$

$$\lambda = a_{11}, a_{22}, a_{33}$$